

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Conducted multi-media environmental sampling and characterization across various field sites, meticulously documenting sample locations and conditions.
- Oversaw drilling operations and contractor activities during site investigations, ensuring adherence to project plans and safety protocols.
- Managed remediation system operations and maintenance tasks, including routine checks, performance monitoring, and minor repairs.
- Prepared technical reports, letters, and memoranda detailing site investigation findings, data evaluation, and remediation recommendations.
- Participated in client and regulatory meetings, presenting technical data and project updates in a clear and concise manner.
- Supported senior team members in client care activities, assisting with proposal development and project planning documentation.
- Analyzed groundwater data using **R** and **Python (pandas, NumPy)** to identify contaminant plumes and assess remediation effectiveness, reducing data processing time by **30%**.

SKILLS

Environmental Science – Hydrogeology, Contaminated Land Assessment, Groundwater Assessment, Remediation, Site Investigation, Environmental Sampling, Characterisation

Engineering & Consulting – Project Management, Budget Management, Technical Reporting, Work Plan Development, Proposal Writing, Client Care, Regulatory Compliance

Fieldwork & Operations – Drilling Oversight, Contractor Oversight, Remediation System Operations & Maintenance, Field Inspections

Data Analysis & Tools – R, Python, pandas, NumPy, SQL, Power BI, Tableau

Software & Systems – Jira, Confluence, Microsoft Office Suite

PROJECTS

Groundwater Contaminant Plume Analysis and Visualization

- Developed Python scripts to process and analyze large datasets of groundwater monitoring well data, identifying trends and spatial distributions of contaminants.
- Created interactive **Power BI** dashboards to visualize contaminant plumes, well locations, and historical concentration data for stakeholder review.
- Applied statistical methods, including regression analysis, to predict contaminant migration pathways and inform remediation strategy.

Site Remediation Operations Dashboard

- Engineered a system using **Java Spring Boot** and **PostgreSQL** to track the operational status and maintenance logs for multiple remediation systems across client sites.
- Implemented REST APIs to allow for real-time data ingestion from field sensors and provide status updates to operations teams.
- Wrote comprehensive unit tests using **JUnit**, achieving over **90%** code coverage for critical service components.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)